

CO3: Manipulate cybersecurity and digital forensic tools or algorithms to identify, investigate, and analyze cyber threats, incidents, and security vulnerabilities.

Assignment Title

[CO3,PO10] RSA Encryption and Decryption Using Student Roll Number

Instructions:

Let your roll number be **R**.

1. Take the **last two digits** of your roll number as the plaintext message (M).
2. Select (p) and (q) according to the last digit of your roll number:

Last Digit of Roll	p	q	e
0,1,2	11	13	7
3,4,5	13	17	5
6,7	17	19	5
8,9	19	23	7

3. Calculate:
 - $n = p \times q$
 - $z = (p-1) \times (q-1)$
 - Private key (d) = $(1+kz)/e$
4. Construct:
 - Public Key ((e,n))
 - Private Key ((d,n))
5. Encrypt the message: $C = M^e \bmod n$
6. Decrypt the ciphertext: $M = C^d \bmod n$
7. Verify that the decrypted message matches the original plaintext.

Mark Distribution (10 Marks)

Task	Marks
Deriving input from roll number correctly for p, q, e	1
Calculate private key correctly	2
Encryption process	3
Decryption and verification	4

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Example

For roll **223071019**:

- Last digit = 9
- ($p = 19$), ($q = 23$), ($e = 7$)
- Plaintext ($M = 19$) (last two digits)